2. THE SIMPLEST VAN DER CORPUT ESTIMATES

2.1 ESTIMATES USING FIRST AND SECOND DERIVATIVES

In the next two chapters, we will obtain upper bounds for sums of the form

$$S = \sum_{n \in I} e(f(n)), \tag{2.1.1}$$

where $I = (a, b]$. (We assume throughout this chapter that a and b are integers.) In this chapter, we obtain the simplest such upper bounds.

From the triangle inequality, we see that

$$|S| \leq \sum_{n \in I} |e(f(n)| = |I| = b - a. \tag{2.1.2}$$

This estimate (henceforth called the *trivial estimate*) holds with equality if $f(n) = xn + y$ and x is an integer. Thus any improvement on (2.1.2) will require some hypothesis on f.

It is instructive to examine the case $f(n) = xn + y$ more closely. If x is not an integer, then

$$|\sum_{n \in I} e(xn + y)| = \frac{|e(bx) - e(ax)|}{|e(x) - 1|} \leq \frac{1}{|\sin \pi x|}. \tag{2.1.3}$$

An examination of the graphs $y = \sin \pi x$ and $y = 2x$ shows that $\sin \pi x \geq 2x$ for $0 \leq x \leq 1/2$. Now define

$$||x|| = \min\{|x - n| : n \in \mathbf{Z}\}.$$

Since $|\sin \pi x|$ is periodic with period 1 and $\sin \pi(1 - x) = \sin \pi x$, we have

$$|\sin \pi x| \geq 2||x||$$

for all real x. Consequently,

$$|\sum_{n \in I} e(xn + y)| \ll ||x||^{-1}.$$

Our first theorem generalizes this result.

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THEOREM 2.1. *(Kusmin-Landau) If f is continuously differentiable, f' is monotonic, and $\|f'\| \geq \lambda > 0$ on I then*

$$\sum_{n\in I} e(f(n)) \ll \lambda^{-1}.$$

Proof. Since

$$\Big|\sum_{n\in I} e(f(n))\Big| = \Big|\sum_{n\in I} e(-f(n))\Big|,$$

we may assume that f' is increasing. Our hypotheses imply that for some integer k,

$$k + \lambda \leq f'(n) \leq k + 1 - \lambda.$$

Now

$$\sum_{n\in I} e(f(n)) = \sum_{n\in I} e(f(n) - kn),$$

so we may assume that $\lambda \leq f'(n) \leq 1 - \lambda$.

Define $g(n) = f(n+1) - f(n)$. From the mean value theorem, we see that there is some x_n such that $n \leq x_n \leq n+1$ and $g(n) = f'(x_n)$. Consequently, g is increasing and $\lambda \leq g(n) \leq 1 - \lambda$. We write

$$e(f(n)) = \frac{e(f(n)) - e(f(n+1))}{1 - e(g(n))} = \{e(f(n)) - e(f(n+1))\}C_n,$$

where $C_n = \frac{1}{2}(1 + i\cot \pi g(n))$. Therefore

$$\begin{aligned}\sum_{n\in I} e(f(n)) &= \sum_{n=a+1}^{b-1} \{e(f(n)) - e(f(n+1)\}C_n + e(f(b)) \\ &= \sum_{n=a+2}^{b-1} e(f(n))(C_n - C_{n-1}) + e(f(a+1))C_{a+1} + e(f(b))(1 - C_{b-1}),\end{aligned}$$

and thus

$$\Big|\sum_{n\in I} e(f(n))\Big| \leq \frac{1}{2} \sum_{n=a+2}^{b-1} |\cot(\pi g(n-1)) - \cot(\pi g(n))| + |C_{a+1}| + |1 - C_{b-1}|.$$

Since $\cot \pi g(n)$ is a decreasing function, the absolute value bars in the right-hand sum may be removed. The remaining sum telescopes, so

$$\Big|\sum_{n\in I} e(f(n))\Big| \leq \frac{1}{2}\{\cot(\pi g(a+1)) - \cot(\pi g(b-1))\} + |C_{a+1}| + |1 - C_{b-1}|.$$

The theorem follows by using the bound $|\cot \pi x| \ll \|x\|^{-1}$.

It is clear from (2.1.3) that Theorem 2.1 is sharp. On the other hand, the hypothesis $\|f'\| \geq \lambda$ is rather restrictive. Our next theorem relaxes this hypothesis.

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THEOREM 2.2. *(van der Corput) Suppose that f is a real valued function with two continuous derivatives on I. Suppose also that there is some $\lambda > 0$ and some $\alpha \geq 1$ such that*

$$\lambda \leq |f''(x)| \leq \alpha\lambda$$

on I. Then

$$\sum_{n\in I} e(f(n)) \ll \alpha|I|\lambda^{1/2} + \lambda^{-1/2}.$$

Proof. Let $\delta(< 1/2)$ be a parameter to be chosen later. From the hypothesis, it follows that I can be split into $\leq \alpha|I|\lambda+2$ intervals on which $||f'|| \geq \delta$, and $\leq \alpha|I|\lambda+1$ other intervals, each of length $\leq 2\delta/\lambda$. We apply Theorem 2.1 to the former set of intervals, and the trivial estimate to the latter. We get

$$\sum_{n\in I} e(f(n)) \ll (\alpha|I|\lambda + 1)(1/\delta + \delta/\lambda + 1).$$

We choose $\delta = \lambda^{1/2}$; this proves the desired result if $\lambda \leq 1/4$. If $\lambda > 1/4$, the result follows from the trivial estimate.

2.2 SOME SIMPLE INEQUALITIES

In this section, we will derive a couple of results that we shall be using throughout this book. Both are simple but, nonetheless, useful.

LEMMA 2.3. *If $X_1, \ldots, X_k$ are positive numbers and $a_1, \ldots, a_k$ are non-negative numbers such that $a_1 + \ldots + a_k = 1$, then*

$$\min(X_1, \ldots, X_k) \leq X_1^{a_1} \ldots X_k^{a_k} \leq \max(X_1, \ldots, X_k).$$

The proof is obvious. We shall often refer to this as the *convexity principle.*

Our second Lemma generalizes the following well known principle. Suppose we have an estimate of the form

$$U \ll AH^a + BH^{-b},$$

where A, B, a, and b are positive constants and H is at our disposal. We obviously want to take H so as to minimize the right hand side. By choosing H to satisfy $AH^a = BH^{-b}$, we get

$$U \ll (A^b B^a)^{1/(a+b)},$$

and this is best possible apart from the value of the implied constant. To generalize this principle, we prove

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LEMMA 2.4. *Suppose that*

$$L(H) = \sum_{i=1}^{m} A_i H^{a_i} + \sum_{j=1}^{n} B_j H^{-b_j},$$

where A_i, B_j, a_i, and b_j are positive. Assume that $H_1 \le H_2$. Then there is some H with $H_1 \le H \le H_2$ and

$$L(H) \ll \sum_{i=1}^{m}\sum_{j=1}^{n} (A_i^{b_j} B_j^{a_i})^{1/(a_i+b_j)} + \sum_{i=1}^{m} A_i H_1^{a_i} + \sum_{j=1}^{n} B_j H_2^{-b_j}.$$

The implied constants depend only on m and n.

Proof. Define

$$L_+(H) = \max(A_1 H^{a_1}, \dots, A_m H^{a_m})$$

and

$$L_-(H) = \max(B_1 H^{-b_1}, \dots, B_n H^{-b_n}).$$

Now $L \le mL_+ + nL_-$, so it suffices to bound L_+ and L_-. We observe that L_+ is a strictly increasing continuous function, $L_+(0) = 0$, and $L_+(\infty) = \infty$. Similarly L_- is a strictly decreasing continuous function, $L_-(0) = \infty$, and $L_-(\infty) = 0$. Therefore there is a unique H_0 such that $L_+(H_0) = L_-(H_0)$. We distinguish three cases: (a) $H_1 \le H_0 \le H_2$, (b) $H_0 < H_1$, (c) $H_0 > H_2$.

If $H_1 \le H_0 \le H_2$ then there is some i and some j such that

$$L_+(H_0) = A_i H_0^{a_i} = L_-(H_0) = B_j H_0^{-b_j}.$$

Consequently

$$L_+(H_0) = L_-(H_0) = (A_i^{b_j} B_j^{a_i})^{1/(a_i+b_j)}.$$

If $H_0 < H_1$ then

$$L_-(H_1) < L_+(H_1) \le \sum_{i=1}^{m} A_i H_1^{a_i}.$$

If $H_0 > H_2$ then

$$L_+(H_2) < L_-(H_2) \le \sum_{j=1}^{n} B_j H_2^{-b_j}.$$

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2.3 THE WEYL-VAN DER CORPUT INEQUALITY

Theorem 2.2 has the disadvantage of being worse than the trivial estimate if $\lambda > 1$. This limitation can be overcome if we replace f by a function with smaller derivatives; the next lemma allows us to make such a replacement.

LEMMA 2.5. *(Weyl-van der Corput) Suppose $\xi(n)$ is a complex valued function such that $\xi(n) = 0$ if $n \notin I$. If H is a positive integer then*

$$|\sum_n \xi(n)|^2 \le \frac{|I|+H}{H} \sum_{|h|<H} \left(1 - \frac{|h|}{H}\right) \sum_n \xi(n)\overline{\xi(n-h)}.$$

Proof. Observe that

$$H \sum_n \xi(n) = \sum_{k=1}^{H} \sum_n \xi(n+k) = \sum_n \sum_{k=1}^{H} \xi(n+k).$$

The inner sum is empty unless $a - H < n \le b - 1$. By Cauchy's inequality,

$$\begin{aligned} H^2 |\sum_n \xi(n)|^2 &\le (|I|+H) \sum_n |\sum_{k=1}^{H} \xi(n+k)|^2 \\ &= (|I|+H) \sum_{k=1}^{H} \sum_{l=1}^{H} \sum_n \overline{\xi(n+k)}\xi(n+l). \end{aligned}$$

We collect together the terms with $l - k = h$ to get the result.

For applications of Lemma 2.5 in this chapter, we will take

$$\xi(n) = \begin{cases} e(f(n)) & \text{if } n \in I; \\ 0 & \text{otherwise.} \end{cases}$$

Let S be as in (2.1.1) and define

$$S_1(h) = \sum_{n \in I(h)} e(f(n+h) - f(n)), \tag{2.3.1}$$

where $I(h) = \{n : n \in I \text{ and } n + h \in I\}$. We see that

$$|S|^2 \le \frac{(|I|+H)}{H} \sum_{|h|<H} |S_1(h)|.$$

If we also assume that $H \le |I|$ then

$$|S|^2 \le \frac{2|I|}{H} \sum_{|h|<H} |S_1(h)|. \tag{2.3.2}$$

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Up to now, we have assumed that H is an integer. If we allow H to be an arbitrary positive number and set $H_0 = [H] + 1$, we get

$$|S|^2 \le \frac{2|I|}{H_0} \sum_{|h|<H_0} |S_1(h)| \le \frac{2|I|}{H} \sum_{|h|\le H} |S_1(h)|. \tag{2.3.3}$$

Furthermore, if $h > 0$ then

$$S_1(h) = \sum_{a<n\le b-h} e(f(n+h) - f(n))$$

and

$$S_1(-h) = \sum_{a+h<n\le b} e(f(n-h) - f(n)).$$

By making a change of variable, we see that $S_1(-h) = \overline{S_1(h)}$. Therefore

$$|S|^2 \le \frac{2|I|^2}{H} + \frac{4|I|}{H} \sum_{1\le h\le H} |S_1(h)|. \tag{2.3.4}$$

We make two further comments that will be relevant in the next section. Suppose I' is an interval that contains I. Then the proof of (2.3.4) may be easily modified to yield

$$|S|^2 \le \frac{2|I'|^2}{H} + \frac{4|I'|}{H} \sum_{1\le h\le H} |S_1(h)|. \tag{2.3.5}$$

provided only that $H \le |I'|$. There are times (cf. Theorem 2.8) when this weaker condition on H is convenient. The second comment is that the function $f(n+h)-f(n)$ may be written as

$$\int_0^1 \frac{\partial}{\partial t} f(n+ht)\, dt.$$

Such integral representations are convenient when Lemma 2.5 is iterated.

By combining (2.3.4) and Theorem 2.2, we prove

THEOREM 2.6. *Suppose that f is a real valued function with three continuous derivatives on I. Suppose also that for some $\lambda > 0$ and for some $\alpha \ge 1$,*

$$\lambda \le |f^{(3)}(x)| \le \alpha\lambda$$

on I. Then

$$S \ll |I|\lambda^{1/6}\alpha^{1/3} + |I|^{3/4}\alpha^{1/4} + |I|^{1/4}\lambda^{-1/4}.$$

Proof. It suffices to bound

$$\frac{2|I|^2}{H} + \frac{4|I|}{H} \sum_{1\le h\le H} |S_1(h)|.$$

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If we set $f_1(n;h) = f(n+h) - f(n)$, then

$$f_1''(n;h) = \int_0^1 hf^{(3)}(n+th)\,dt,$$

and so

$$h\lambda \le |f_1''(n;h)| \le \alpha h\lambda.$$

From Theorem 2.2 we see that

$$|S_1(h)| \ll |I|h^{1/2}\alpha\lambda^{1/2} + h^{-1/2}\lambda^{-1/2}.$$

Consequently,

$$|S|^2 \ll |I|^2H^{-1} + |I|^2H^{1/2}\lambda^{1/2}\alpha + |I|H^{-1/2}\lambda^{-1/2}.$$

We apply Lemma 2.4 with $H_2 = |I|, H_1 = 0$ to get

$$|S|^2 \ll |I|^2\lambda^{1/3}\alpha^{2/3} + |I|^{3/2}\alpha^{1/2} + |I| + |I|^{1/2}\lambda^{-1/2}.$$

Because the second term dominates the third, the result follows.

We now give an example which provides a preview of the next section. We shall obtain upper bounds for

$$S = \sum_{N<n\le 2N} e(An^{-\sigma}),$$

where σ is a real number, $\sigma \ne 0, -1$, or -2, and $A > 0$.

In this example, all implied constants are allowed to depend on σ, but they are otherwise absolute. Let $f(n) = An^{-\sigma}$ and $F = AN^{-\sigma}$. By Theorems 2.2 and 2.6,

$$S \ll F^{1/2} + F^{-1/2}N \tag{2.3.6}$$

and

$$S \ll F^{1/6}N^{1/2} + N^{3/4} + F^{-1/4}N. \tag{2.3.7}$$

In fact, these estimates may be sharpened to

$$S \ll F^{1/2} + F^{-1}N \tag{2.3.8}$$

and

$$S \ll F^{1/6}N^{1/2} + F^{-1}N. \tag{2.3.9}$$

To prove (2.3.8), assume first that $\sigma > -1$. If $F \ge (2|\sigma|)^{-1}N$, then $S \ll F^{1/2}$ by (2.3.6). In the contrary case, we can apply the Kusmin-Landau inequality to get $S \ll F^{-1}N$.

If $\sigma \le -1$, we argue in a similar fashion; we apply (2.3.6) if $F \ge 2^\sigma|\sigma|^{-1}N$ and Kusmin-Landau otherwise.

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To prove (2.3.9), we let

$$E_1 = \max(F^{1/2}, F^{-1}N) \text{ and } E_2 = \max(F^{1/6}N^{1/2}, N^{3/4}, F^{-1/4}N).$$

Therefore, $S \ll \min(E_1, E_2)$ and the desired estimate follows immediately unless $E_1 = F^{1/2}$ and $E_2 = N^{3/4}$ or $F^{-1/4}N$. If the first possibility for E_2 holds, then we use Lemma 2.3 (convexity) to get

$$S \ll E_1^{1/3} E_2^{2/3} = F^{1/6} N^{1/2}.$$

Otherwise, we use the same Lemma to get

$$S \ll E_1^{5/9} E_2^{4/9} = F^{1/6} N^{4/9} \leq F^{1/6} N^{1/2}.$$

2.4 ITERATING WEYL-VAN DER CORPUT

We begin by squaring (2.3.4) and using Cauchy's inequality to get

$$|S|^4 \leq \frac{8|I|^4}{H_1^2} + \frac{32|I|^2}{H_1} \sum_{1 \leq h_1 \leq H} |S_1(h_1)|^2.$$

By applying (2.3.5) to $S_1(h_1)$, we get

$$|S|^4 \leq \frac{8|I|^4}{H_1^2} + \frac{64|I|^4}{H_2} + \frac{128|I|^3}{H_1 H_2} \sum_{1 \leq h_1 \leq H_1} \sum_{1 \leq h_2 \leq H_2} |S_2(h_1, h_2)|,$$

provided $H_1 \leq |I|$ and $H_2 \leq |I|$. The summands on the right are defined by

$$S_2(h_1, h_2) = \sum_{n \in I(h_1, h_2)} e(f_2(n; h_1, h_2)),$$

$$\begin{aligned} f_2(n; h_1, h_2) =& f(n + h_1 + h_2) - f(n + h_1) - f(n + h_2) + f(n) \\ =& \int_0^1 \int_0^1 \frac{\partial^2}{\partial t_1 \partial t_2} f(n + t_1 h_1 + t_2 h_2)\, dt_1 dt_2, \end{aligned}$$

and

$$I(h_1, h_2) = \{n : n, n + h_1, n + h_2, \text{ and } n + h_1 + h_2 \in I\}.$$

If we assume that $H_1^2 = H_2$, then we may write

$$|S|^4 \leq 8^3 \Big\{ \frac{|I|^4}{H_2} + \frac{|I|^3}{H_1 H_2} \sum_{1 \leq h_1 \leq H_1} \sum_{1 \leq h_2 \leq H_2} |S_2(h_1, h_2)| \Big\}.$$

This argument may be continued, and a straightforward induction argument yields

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LEMMA 2.7. *Let q be a positive integer, and let $Q = 2^q$. If $H \le |I|$, $H_q = H$, $H_{q-1} = H^{1/2}, \dots$, and $H_1 = H^{2/Q}$, then*

$$|S|^q \le 8^{Q-1}\Big\{\frac{|I|^Q}{H} + \frac{|I|^{Q-1}}{H_1 \dots H_q} \sum_{1\le h_1\le H_1} \cdots \sum_{1\le h_q\le H_q} |S_q(\mathbf{h})|\Big\},$$

where $\mathbf{h} = (h_1, h_2, \dots h_q)$,

$$S_q(\mathbf{h}) = \sum_{n\in I(\mathbf{h})} e(f_q(n;\mathbf{h})),$$

$$f_q(n;\mathbf{h}) = \int_0^1 \cdots \int_0^1 \frac{\partial^q}{\partial t_1 \partial t_2 \dots \partial t_q} f(n + \mathbf{h}\cdot\mathbf{t})\, dt_1 \dots dt_q,$$

$$\mathbf{t} = (t_1, \dots, t_q), \text{ and } I(\mathbf{h}) = (a, b - h_1 - h_2 - \dots - h_q].$$

We combine Lemma 2.7 with Theorem 2.2 to get

THEOREM 2.8. *Let q be a positive integer. Suppose that f is a real valued function with $q+2$ continuous derivatives on I. Suppose also that for some $\lambda > 0$ and for some $\alpha \ge 1$,*

$$\lambda \le |f^{(q+2)}(x)| \le \alpha\lambda$$

on I. Let $Q = 2^q$. Then

$$S \ll |I|(\alpha^2\lambda)^{1/(4Q-2)} + |I|^{1-1/2Q}\alpha^{1/2Q} + |I|^{1-2/Q+1/Q^2}\lambda^{-1/2Q}.$$

The implied constant is absolute.

Proof. The function $f_q(n;\mathbf{h})$ of the previous lemma is equal to

$$\int_0^1 \cdots \int_0^1 h_1 \dots h_q f^{(q)}(n + \mathbf{h}\cdot\mathbf{t})\, dt_1 \dots dt_q;$$

therefore,

$$h_1 h_2 \dots h_q \lambda \le |f_q''(n;\mathbf{h})| \le h_1 h_2 \dots h_q \alpha\lambda.$$

By Theorem 2.2,

$$|S_q(\mathbf{h})| \ll |I|\alpha(h_1 h_2 \dots h_q\lambda)^{1/2} + (h_1 h_2 \dots h_q\lambda)^{-1/2}.$$

We now sum over $\mathbf{h}$. It is useful to note that

$$\sum_{1\le h\le H} h^{-1/2} \le \sum_{1\le h\le H} \int_{h-1}^{h} t^{-1/2}\, dt = \int_0^H t^{-1/2}\, dt = 2H^{1/2};$$

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whence

$$(H_1 \dots H_q)^{-1} \sum_{\mathbf{h}} (h_1 \dots h_q)^{-1/2} \le 2^q (H_1 \dots H_q)^{-1/2}.$$

Similarly,

$$\sum_{1 \le h \le H} h^{1/2} \le H^{1/2} \sum_{1 \le h \le H} 1 \le H^{3/2},$$

and

$$(H_1 \dots H_q)^{-1} \sum_{\mathbf{h}} (h_1 \dots h_q)^{1/2} \le (H_1 \dots H_q)^{1/2}.$$

All together, we get

$$|S|^Q \ll (2^q 8^Q) |I|^Q (H^{-1} + H^{1-1/Q} \alpha \lambda^{1/2} + H^{-1+1/Q} |I|^{-1} \lambda^{-1/2}).$$

We apply Lemma 2.4 to get

$$|S|^Q \ll (2^q 8^Q) |I|^Q \{(\alpha^2 \lambda)^{Q/(4Q-2)} + |I|^{-1/2} \alpha^{1/2} + |I|^{-2+1/Q} \lambda^{-1/2}\}.$$

Taking Qth roots gives the result. Note that $(2^q 8^Q)^{1/Q}$ is bounded, so the implied constant is absolute.

We saw in an example in the previous section how an estimate obtained from Theorem 2.6 can be improved if there is sufficient information about f', f'', and f'''. Our next theorem carries this idea further.

THEOREM 2.9. *Let $q \ge 0$ be an integer. Suppose that f has $q+2$ continuous derivatives on I, and that $I \subseteq (N, 2N]$. Assume also that there is some constant F such that*

$$|f^{(r)}(x)| \approx F N^{-r} \tag{2.4.1}$$

for $r = 1, \dots, q+2$. Then

$$S \ll F^{1/(4Q-2)} N^{1-(q+2)/(4Q-2)} + F^{-1} N.$$

The implied constant depends only upon the implied constants in (2.4.1).

Proof. By (2.4.1) when $r = 1$, we see that there are constants c_1 and c_2 such that

$$c_1 F N^{-1} \le |f'(n)| \le c_2 F N^{-1}.$$

If $c_2 F N^{-1} \le 1/2$, then we can use Theorem 2.1 (Kusmin-Landau) to get $S \ll F^{-1} N$. In the contrary case, Theorem 2.2 yields $S \ll F^{1/2}$. In any case,

$$S \ll F^{1/2} + F^{-1} N,$$

which is the desired result for $q = 0$.

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We complete the proof by induction on q. Assume the theorem is true with q replaced by $q-1$. From the induction hypothesis and Theorem 2.8, we see that $S \ll \min(E_1, E_2)$,where

$$E_1 = \max(F^{1/(2Q-2)}N^{1-(q+1)/(2Q-2)}, F^{-1}N) = \max(M_1, M_2),$$

say, and

$$\begin{aligned} E_2 &= \max(F^{1/(4Q-2)}N^{1-(q+2)/(4Q-2)}, N^{1-1/2Q}, F^{-1/2Q}N^{1+(q-2)/2Q+1/Q^2}) \\ &= \max(M_3, M_4, M_5), \end{aligned}$$

say. The result follows immediately unless we have one of the following cases: (i) $E_1 = M_1, E_2 = M_4$, (ii) $E_1 = M_1, E_2 = M_5$.

In case (i), we get the result by using

$$\min(M_1, M_4) \leq (M_1^{2Q-2}M_4^{2Q})^{1/(4Q-2)}.$$

In case (ii), we use

$$\min(M_1, M_5) \leq M_1^a M_5^b,$$

where

$$a = \frac{(2Q-2)(6Q-2)}{(4Q-2)^2}, \quad b = \frac{4Q^2}{(4Q-2)^2},$$

and the result follows again.

2.5 UPPER BOUNDS FOR THE RIEMANN ZETA-FUNCTION

In this section, we will use $s = \sigma + it$ to denote a complex variable.

Riemann's zeta-function is defined as

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$$

for $\sigma > 1$. This function may be continued to function which is analytic in the whole plane except for a simple pole at $s = 1$. For our purposes, we need this continuation only for $\sigma > 0$; we do this in the course of proving Lemma 2.11.

Here we shall bound $\zeta(s)$ in the range $1/2 \leq \sigma \leq 1$ and $t \geq 3$. The latter restriction allows us to avoid the pole at $s = 1$ and to regard $\log t$ and $\log\log t$ as positive quantities.

Our first two lemmas reduce the problem to estimates of finite exponential sums.

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LEMMA 2.10. *If* $1/2 \le \sigma \le 1$ *and* $N \le M$ *then*

$$\sum_{N<n\le M} n^{-\sigma-it} \ll N^{-\sigma} \max_{N<u\le M} \Big|\sum_{N<n\le u} n^{it}\Big|.$$

Proof. Let

$$S(u) = \sum_{N<n\le u} n^{-it}.$$

Then

$$\begin{aligned}\sum_{N<n\le M} n^{-\sigma-it} &= \int_N^M u^{-\sigma}\, dS(u) = S(M)M^{-\sigma} + \sigma\int_N^M S(u)u^{-\sigma-1}\, du \\ &\ll N^{-\sigma} \max_{N<u\le M} |S(u)|,\end{aligned}$$

and the desired result follows.

LEMMA 2.11. *If* $1/2 \le \sigma \le 1$ *and* $t \ge 3$ *then*

$$|\zeta(\sigma+it)| \ll \Big|\sum_{n\le t} n^{-\sigma-it}\Big| + t^{1-2\sigma}\log t.$$

Proof. If $\sigma > 1$ and $M \ge 1$ then

$$\zeta(s) = \sum_{n\le M} n^{-s} + \int_M^\infty u^{-s}\, d[u] = \sum_{n\le M} n^{-s} + \frac{M^{1-s}}{s-1} + s\int_M^\infty \frac{[u]-u}{u^{s+1}}\, du.$$

The last integral converges for $\sigma > 0$, so this gives an analytic continuation of $\zeta(s)$ to the region $\sigma > 0, s \ne 1$. We set $M = t^2$ and use the inequality $|[u]-u| \le 1$ to get

$$\zeta(s) = \sum_{n\le t^2} n^{-s} + O(t^{1-2\sigma}).$$

The sum over the range $t < n \le t^2$ may be divided into $\ll \log t$ subsums of the form

$$\sum_{N<n\le N_1} n^{-\sigma-it}.$$

where $N_1 = \min(2N, t^2)$. From Lemma 2.10 and Theorem 2.1 (Kusmin-Landau), we see that each of the above subsums is $\ll N^{1-\sigma}t^{-1} \ll t^{1-2\sigma}$, and the result follows.

THEOREM 2.12. *Let* $q \ge 1$ *be an integer,* $Q = 2^q$, *and* $\sigma(q) = 1 - (q+2)/(4Q-2)$. *If* $t \ge 3$ *then*

$$\zeta(\sigma(q)+it) \ll t^{1/(4Q-2)}\log t.$$

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In particular,

$$\zeta(1/2+it) \ll t^{1/6}\log t,$$

and

$$\zeta(5/7+it) \ll t^{1/14}\log t.$$

Proof. By the previous two lemmas, it suffices to show that

$$\sum_{N<n\le M} n^{it} \ll N^{\sigma(q)}t^{1/(4Q-2)}$$

whenever $M \le 2N$ and $N \le t$. This, however, follows immediately from Theorem 2.9 with $f(t) = -(t\log n)/2\pi$ and $F = t$.

Since we appealed to Theorem 2.9, our estimate in the previous theorem depends on q. We could, however, remove such dependence by arguing more carefully from Theorem 2.8. An example of such an argument appears in Theorem 2.14.

Our next result uses the convexity principle to "interpolate" between $\sigma(q)$ and $\sigma(q+1)$.

COROLLARY 2.13. *Let q, Q, and $\sigma(q)$ be as in the previous theorem. If $\sigma(q) \le \sigma \le \sigma(q+1)$ and $t \ge 3$ then*

$$\zeta(\sigma+it) \ll t^{\beta(\sigma)}\log t,$$

where

$$\beta(\sigma) = \frac{\sigma(q+1)-\sigma}{\sigma(q+1)-\sigma(q)}\cdot\frac{1}{4Q-2} + \frac{\sigma-\sigma(q)}{\sigma(q+1)-\sigma(q)}\cdot\frac{1}{8Q-2}.$$

Proof. If $M \le 2N$ and $N \le t$ then

$$\sum_{N<n\le M} n^{-it} \ll \min(E_1, E_2),$$

where

$$E_1 = N^{\sigma(q)}t^{1/(4Q-2)}, E_2 = N^{\sigma(q+1)}t^{1/(8Q-2)}.$$

Now we use $\min(E_1, E_2) \ll E_1^a E_2^b$, where a and b are chosen so that $a+b=1$ and $a\sigma(q)+b\sigma(q+1)=\sigma$. In other words,

$$a = \frac{\sigma(q+1)-\sigma}{\sigma(q+1)-\sigma(q)}\ ,\ b = \frac{\sigma-\sigma(q)}{\sigma(q+1)-\sigma(q)}.$$

It follows that

$$\sum_{N<n\le M} n^{it} \ll t^{\beta(\sigma)}N^{\sigma},$$

and the desired result follows from Lemmas 2.10 and 2.11.

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THEOREM 2.14. *If $t \geq 3$ then*

$$\zeta(1+it) \ll \frac{\log t}{\log\log t}.$$

Proof. From Lemma 2.11, we have

$$\zeta(1+it) = \sum_{n \leq R} n^{-1-it} + \sum_{R < n \leq t} n^{-1-it} + O(t^{-1}\log t). \tag{2.5.1}$$

The first sum is trivially $\ll \log R$. Our approach is to make the second sum $\ll 1$ for R as large as possible.

Suppose $R \leq N \leq t$ and $M \leq 2N$. Let

$$S = \sum_{N < n \leq M} n^{-it} = \sum_{N < n \leq M} e(f(n)),$$

say. For $q \geq 1$, we have

$$|f^{(q+2)}(n)| = \frac{(q+1)!t}{2\pi n^{q+2}}.$$

Therefore, the hypotheses of Theorem 2.8 are satisfied with

$$\lambda = \frac{(q+1)!t}{2\pi(2N)^{q+2}}\ ,\ \alpha = 2^{q+2}.$$

Consequently,

$$S \ll t^{1/(4Q-2)}N^{1-(q+2)/(4Q-2)} + N^{1-1/2Q} + t^{-1/2Q}N^{1+(q-2)/2Q+1/Q^2},$$

and this estimate is uniform in q. We also have $S \ll t^{1/2}$ from Theorem 2.2. We can now repeat the argument of Theorem 2.9 to obtain

$$S \ll t^{1/(4Q-2)}N^{1-(q+2)/(4Q-2)}. \tag{2.5.2}$$

This estimate is valid for $q \geq 0$, and the implied constant does not depend on q.

We use a splitting argument, Lemma 2.10, and (2.5.2) to get

$$\sum_{R < n \leq t} n^{-1-it} \ll t^{1/(4Q-2)}R^{-(q+2)/(4Q-2)}\log t \tag{2.5.3}$$

for $q \geq 0$. To make an appropriate choice of q, we observe that the left hand side of (2.5.3) is

$$\exp\left(\log\log t + \frac{\log t - (q+2)\log R}{4Q-2}\right),$$

and this is $\ll 1$ if

$$(\log\log t)(4Q-2) + \log t \leq (q+2)\log R. \tag{2.5.4}$$

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We now let $R = \exp(\log t/c\log\log t)$ and $q = [b\log t/\log R] - 2$, where b and c will be chosen later. Then

$$4Q = \exp((q+2)\log 2) \le \exp(b\log t\log 2/\log R) = (\log t)^{bc\log 2}.$$

We choose $b = 3/2$ and $c = 1/(2\log 2)$. Then $4Q \le (\log t)^{3/4}$, so

$$(\log\log t)(4Q-2) + \log t \le \log t + (\log\log t)(\log t)^{3/4} \le b\log t \le (q+2)\log R$$

for t sufficiently large. Thus (2.5.4) is satisfied. From (2.5.1) and our choice of R we get

$$\zeta(1+it) \ll \log R \ll \frac{\log t}{\log\log t}.$$

2.6 NOTES

Theorem 2.1, while sufficient for our purposes, is a weak version of what is usually referred to as the Kusmin-Landau inequality. Suppose f and θ are such that

$$0 < \theta \le f(2) - f(1) \le \ldots \le f(N) - f(N-1) < 1 - \theta$$

and $S = \sum_{n=1}^{N} e(f(n))$. Van der Corput (1921) proved that $S \ll 1/\theta$. Kusmin (1927) sharpened this to $|S| \le 1/\theta$, and Landau (1928) gave the estimate $|S| \le \cot(\pi\theta/2)$. Elementary proofs of all of the these results can be found in a paper of Mordell (1958). The Kusmin-Landau inequality was rediscovered by Herzog and Piranian (1949) in some work on sets of convergence of Taylor series. We have essentially followed them in our proof.

Lemma 2.4, in the special case $H_1 = 0$ and $H_2 = \infty$, is due to van der Corput (1922). Srinivasan (1983) first enunciated the general case.

Despite it simplicity and usefulness, the convexity principle has not always been well appreciated by researchers in exponential sums. For example, it does not appear explicitly in the works of van der Corput, Titchmarsh, or Phillips. For a nice illustration of the power of convexity, see the work of Heath-Brown and Iwaniec (1979) on gaps between primes.

Theorem 2.8 should be compared to Theorem 5.13 of Titchmarsh (1951). The main terms are identical, but the secondary terms in our theorem are smaller. The improvement comes from the use of Lemma 2.4.

The estimate $\zeta(1+it) \ll \log t/\log\log t$ was first proved by Weyl (1921). Stronger estimates can be obtained from Vinogradov's mean-value estimates (see Titchmarsh (1951, Chapter 6) and Ivić (1985, Chapter 6)).

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